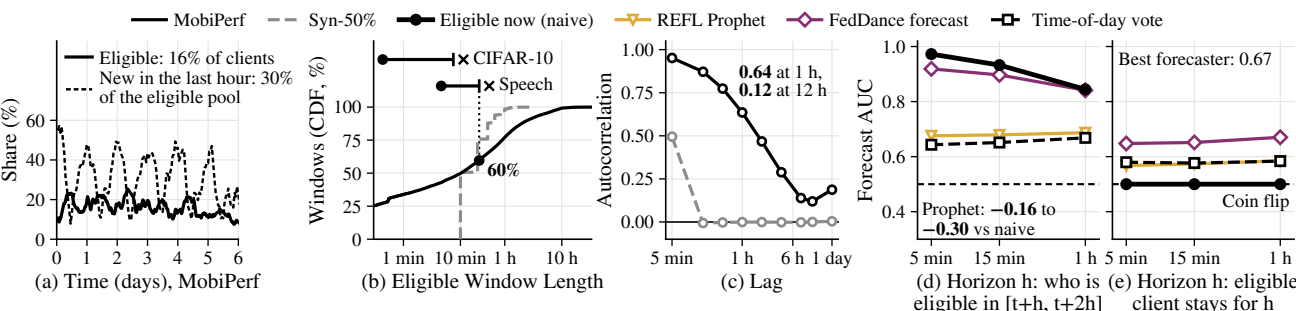
MobiPerf
Syn-50%
Eligible now (naive)
REFL Prophet
FedDance forecast
Time-of-day vote
Eligible: 16% of clients
New in the last hour: 30%
of the eligible pool
Share (%)
0
20
40
60
0 1 2 3 4 5 6
(a) Time (days), MobiPerf
CIFAR-10
Speech
60%
Windows (CDF, %)
0
25
50
75
100
1 min 10 min 1 h 10 h
(b) Eligible Window Length
Autocorrelation
0.00
0.25
0.50
0.75
1.00
0.64 at 1 h,
0.12 at 12 h
5 min 1 h 6 h 1 day
(c) Lag
Forecast AUC
0.4
0.6
0.8
1.0
Prophet: −0.16 to
−0.30 vs naive
5 min 15 min 1 h
(d) Horizon h: who is
eligible in [t+h, t+2h]
Best forecaster: 0.67
Coin flip
5 min 15 min 1 h
(e) Horizon h: eligible
client stays for h